

Ian Douglas Mager

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EDUCATION

Bachelor of Science in Computer Science — May 2026

Texas State University, San Marcos, TX

Relevant Coursework: Machine Learning, Compiler Construction, Database Systems, Robotics, Algorithms & Data Structures

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks & Tools: Jira, FastAPI, PostgreSQL, Supabase, Firebase, Node.js, Git, scikit-learn

PROJECTS

Autonomous Robot Navigation System | *Texas State University — Robotics* Jan 2026 – May 2026

- Collaborated with a 4-person team to design and implement an autonomous navigation system for a TurtleBot3 platform using ROS 2, integrating camera and LiDAR sensor data to enable real-time environment perception.
- Developed and tuned navigation logic responsible for goal-directed path planning, enabling the robot to autonomously interpret a target destination and execute collision-free movement to reach it.
- Delivered a fully functional robot capable of completing all assigned navigation tasks in end-of-semester demonstration, validating the accuracy and reliability of the sensor-fusion and path-planning pipeline.

University Database Management System | *Texas State University — Database Systems* Aug 2025 – Dec 2025

- Identified the need for a structured data management solution for university operations, designing and building a full-stack application to model and query student, course, and faculty data.
- Architected and implemented the entire system independently — including a normalized PostgreSQL schema, RESTful API endpoints with FastAPI, and a cloud-hosted backend using Supabase.
- Delivered a production-ready application with end-to-end data integrity, demonstrating correct relational modeling and live query performance across multiple interconnected university entities.

AI-Powered Daily Puzzle Application | *Personal Project* 2025

- Recognized an opportunity to combine generative AI with gamification, designing and building a full-stack puzzle application that serves unique, AI-generated visual puzzles to users on a daily rotation.
- Engineered the core puzzle generation pipeline in JavaScript, integrating DALL·E for image synthesis and Gemini for dynamic content generation, with Firebase handling authentication, storage, and real-time data.
- Deployed the application on Render with a daily puzzle rotation mechanism, delivering a seamless user experience with fresh, AI-generated content automatically refreshed each day.

ADDITIONAL EXPERIENCE

Compiler Construction | *Texas State University* Spring 2026

- Designed and independently debugged a complete four-phase compiler for a C-like language. Implementing a lexer, YACC/Bison parser producing an AST and symbol table, semantic analyzer, and MIPS assembly code generator

Machine Learning - Kaggle Competition Project | *Texas State University* Spring 2026

- Led model selection and feature engineering for a multi-class irrigation prediction task using scikit-learn, applying cross-validation and regularization strategies to address class imbalance on real-world agricultural data.